Programmable Logic devices -:

PLD -: i) PLD is special type of IC's which can be programmed as per the requirement of user.

ii) The combinational or sequential ckt. using the PLD's is possible

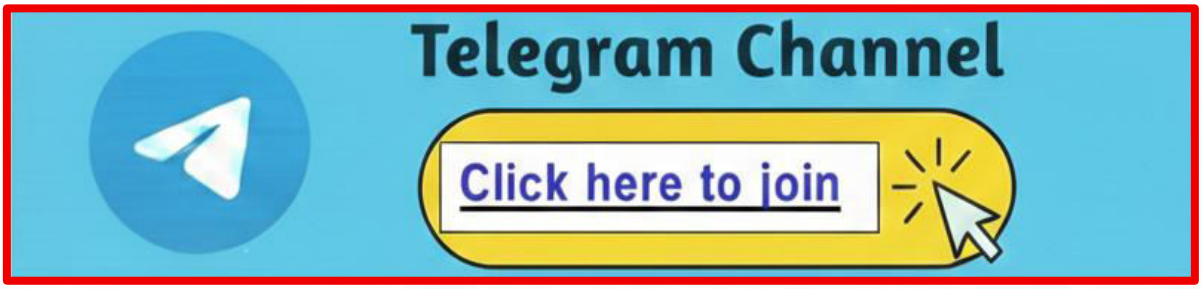
TYPES OF PLD's

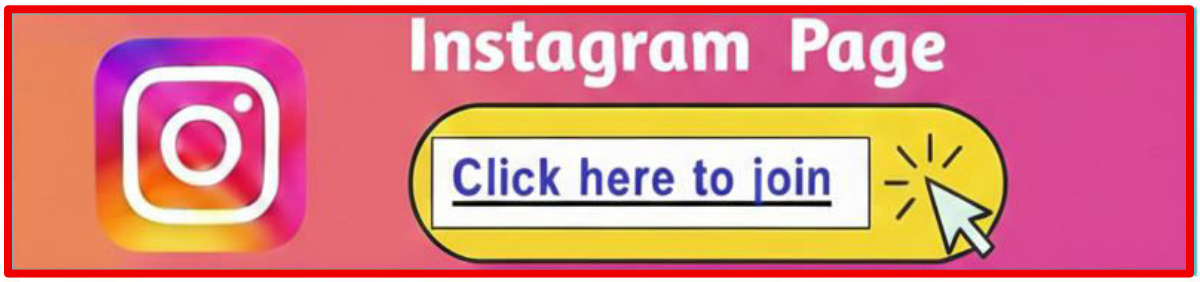
1) Read only Memory (ROM's)
2) PLA's (programmable logic Array (PLA's)
3) programmable Array Logic (PAL's)

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1) ROM used as PLD →

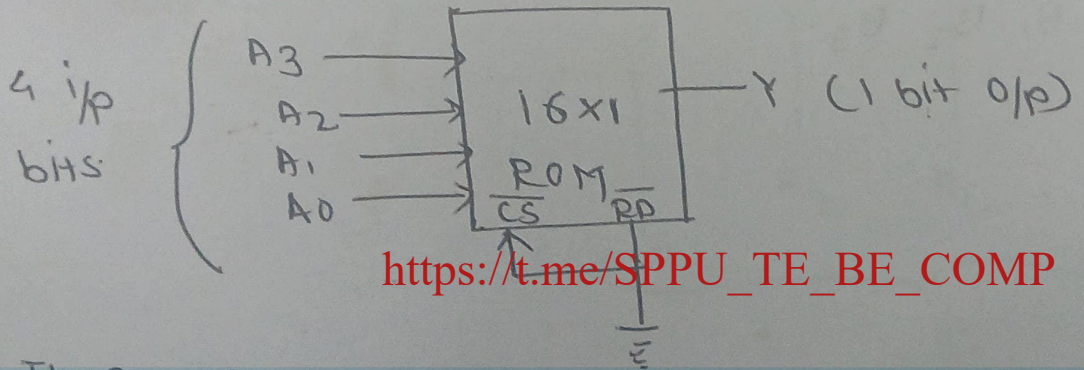
ROM - Read only Memory. ROM size = M x N

P = No of address line to access M. location.

$2^P = M$.

N = data lines.

ROM of 16x1 M = 16 N = 1



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PROM :- eg

Que :- Design a 3-bit gray to binary code converter. Implement it using suitable PROM. Draw this PROM table & logic diagram.

⇒ Truth table

Gray i/p			Binary output		
G_2	G_1	G_0	B_2	B_1	B_0
0	0	0	0	0	0
0	0	1	0	0	1
0	1	1	0	1	0
0	1	0	0	1	1
1	1	0	1	0	0
1	1	1	1	0	1
1	0	1	1	1	0
1	0	0	1	1	1

$B_2 = G_2$

$B_1 = \overline{G_2} \cdot G_1 + G_2 \cdot \overline{G_1}$

$B_0 = \overline{G_2}\,\overline{G_1}\,G_0 + \overline{G_2}\,G_1\,\overline{G_0} + G_2 G_1 G_0 + G_2 \overline{G_1}\,\overline{G_0}$

ROM programming table :-

ROM – 3 i/p variable $2^3 = 8$ memory location. Each location can store 4-bit word. 4 o/p.

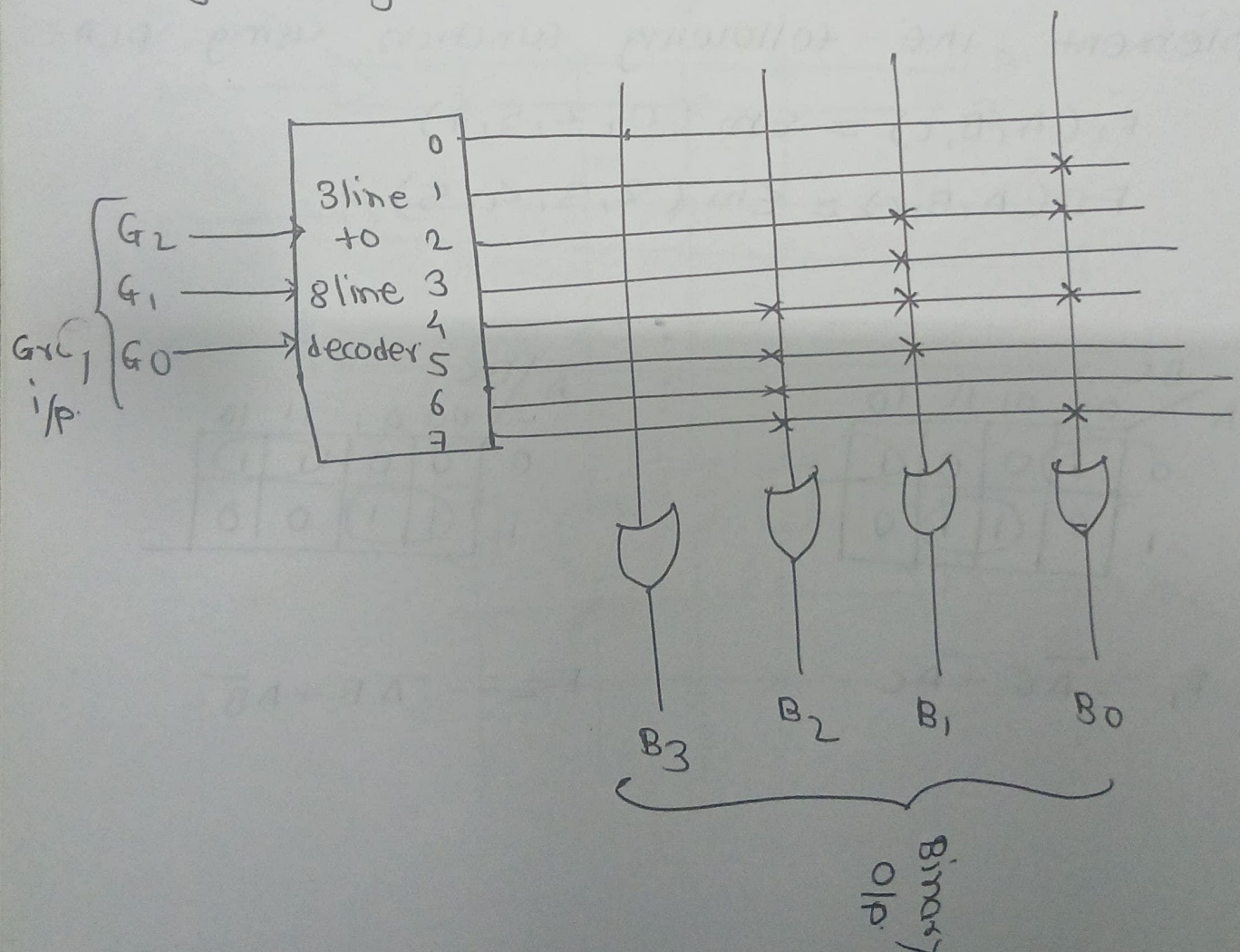
B_0, B_1, B_2, B_3

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ROM programming table

Location number	I/P G_2	G_1	G_0	O/P B_3	B_2	B_1	B_0	
0	0	0	0	0	0	0	0	← Data in location 0
1	0	0	1	0	0	0	1	
2	0	1	0	0	0	1	1	
3	0	1	1	0	0	1	0	
4	1	0	0	0	1	1	1	
5	1	0	1	0	1	1	0	
6	1	1	0	0	1	0	0	
7	1	1	1	0	1	0	1	← Data in location 7

Logic diagram :-



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Programmable Logic Array (PLA) :-

i) PLD generally consists of programmable array of logic gates. Interconnections are made with the array inputs.

ii) Two level AND-OR configuration is used.

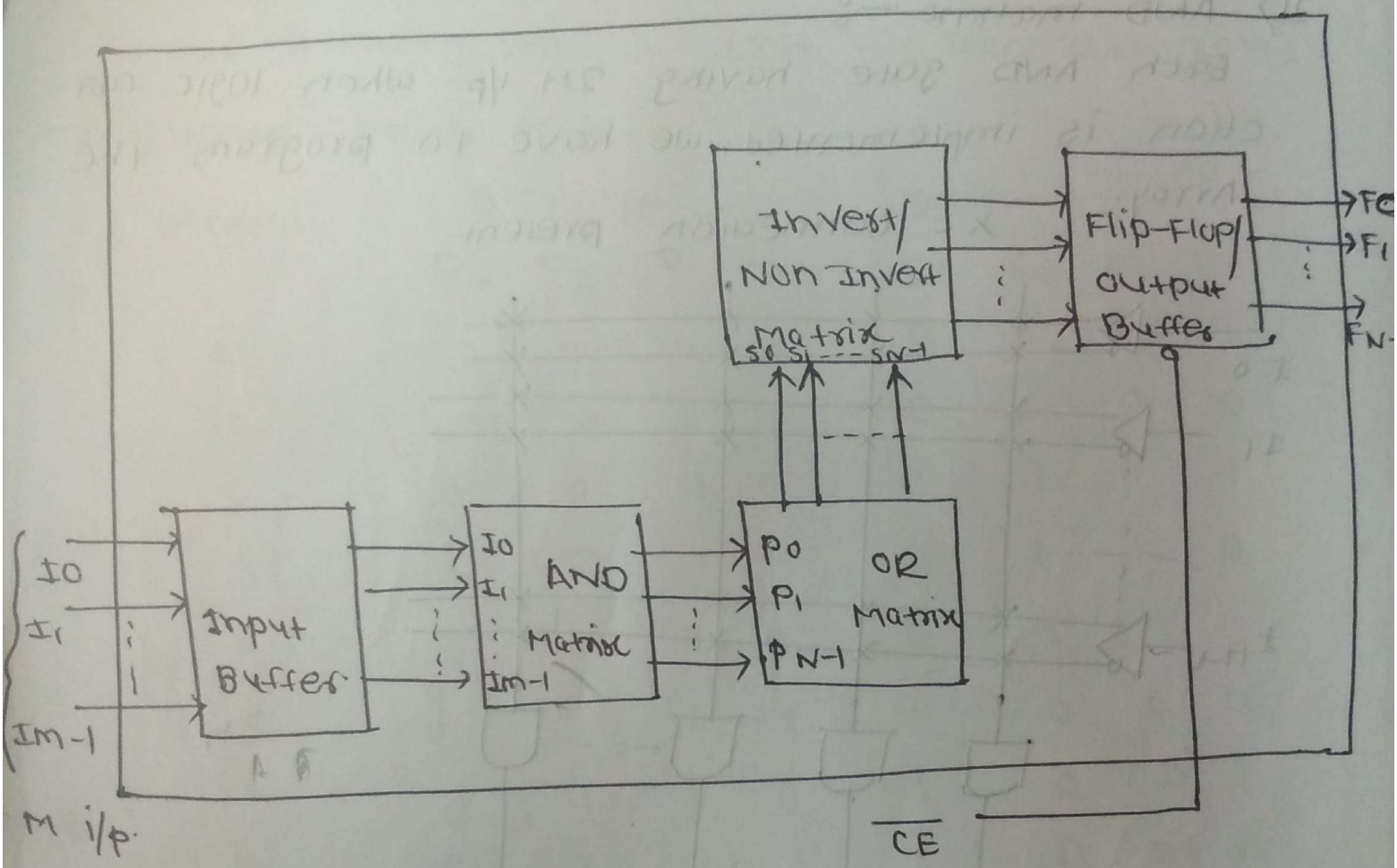


Fig → Block diagram of PLA Device.

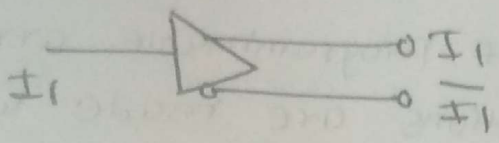
iii) Due to the use of AND matrix followed by OR Matrix, we can use them for the implementation of logic functions in the SOP form.

iv) AND matrix it is wed for implement the product terms in SOP form & the OR matrix is wed for implementing the sum of product term.

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I) Input Buffer :-

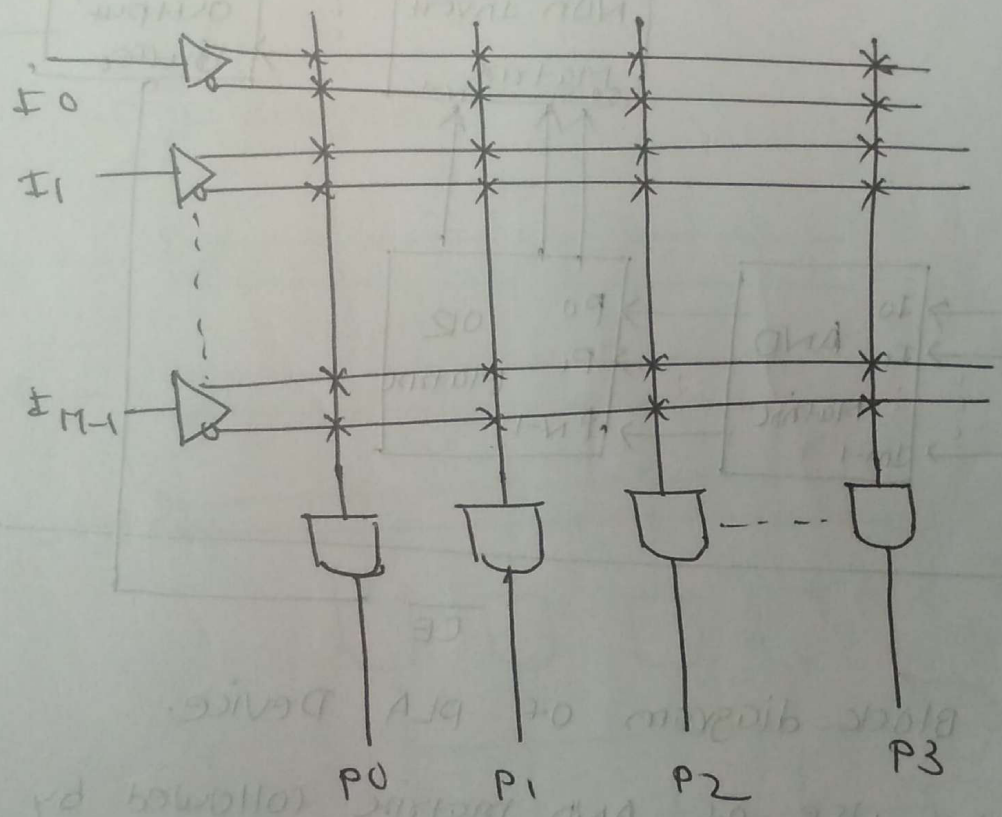


Buffer are of two types inverting & non-Inverting Buffer.

II) AND matrix :-

Each AND gate having 2M i/p. when logic function is implemented, we have to program the Array.

X = connection present.



III) OR Matrix

IV) Invert / Non-invert matrix :-

It is basically programmable Buffer.

V) O/p Buffer :- (Increase current sourcing capability)

$\overline{CE}$ = 1 i/p applied to o/p

$\overline{CE}$ = 0 o/p will be made available.

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Programmable Logic Array (PLA) :-

Both AND & OR programmable

Example :-

1) Implement following function using PLA.

$$f_1(A,B,C) = \Sigma m\,(0,2,5,7)$$

$$f_2(A,B,C) = \Sigma m\,(2,3,4,5)$$

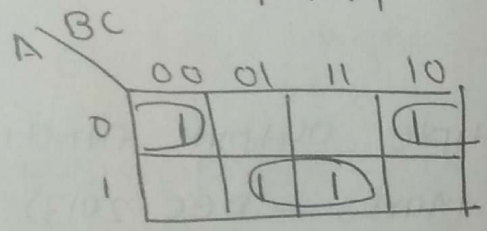
$\Rightarrow$

$$F_1(A,B,C) = \Sigma m\,(0,2,5,7)$$

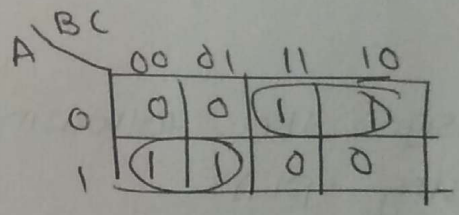
$$f_2(A,B,C) = \Sigma m\,(2,3,4,5)$$

(i) K-map & simplifications

For F_1



For F_2



$F_1 = A\bar{C} + \bar{A}\bar{C}$

(1) (2)

$F_2 = A\bar{B} + \bar{A}B$

(3) (4)

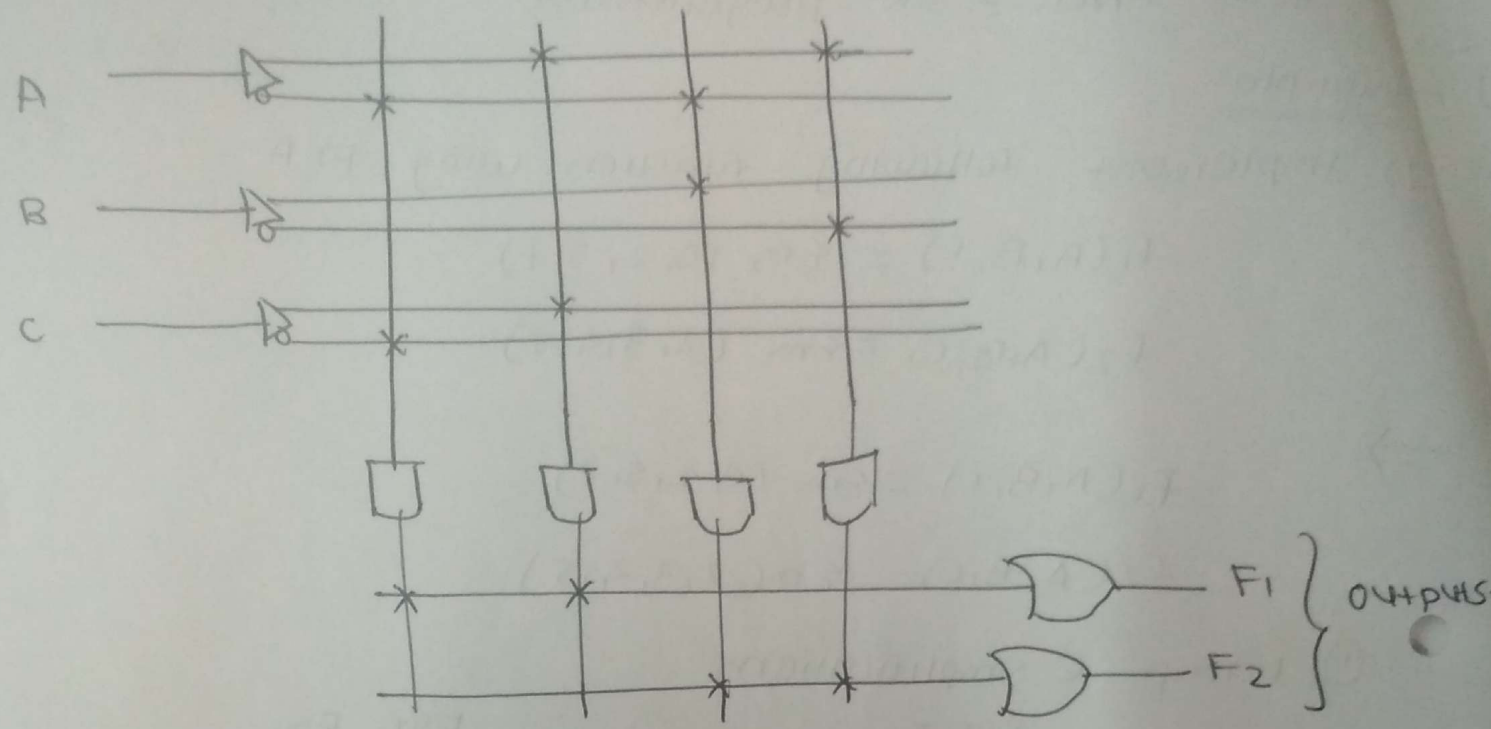
(ii) prepare PLA programming table

product term NO.	product term	I/p A	I/p B	I/p C	Outputs F_1	Outputs F_2
1	$A\bar{C}$	1	–	0	1	–
2	$\bar{A}\bar{C}$	0	–	0	1	–
3	$\bar{A}B$	0	1	–	–	1
4	$A\bar{B}$	1	0	–	–	1

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iii) Implement logic circuit :-



2) Design the following multiple output function using PLA. (7 Marks - Dec 2013)

$F_1(a,b,c,d) = \Sigma m\ (3, 7, 8, 9, 11, 15)$

$F_2(a,b,c,d) = \Sigma m\ (3, 4, 5, 7, 10, 14, 15)$

⇒ i) K-map & simplification

For F_1

AB \ CD	00	01	11	10
00	0	0	1	0
01	0	0	1	0
11	0	0	1	0
10	1	1	1	0

$F_1 = A\bar{B}\bar{C} + CD$

AB \ CD	00	01	11	10
00	0	0	1	0
01	1	1	1	0
11	0	0	1	1
10	0	0	0	1

$F_2 = \bar{A}B\bar{C} + \bar{A}CD + ABC + AC\bar{D}$

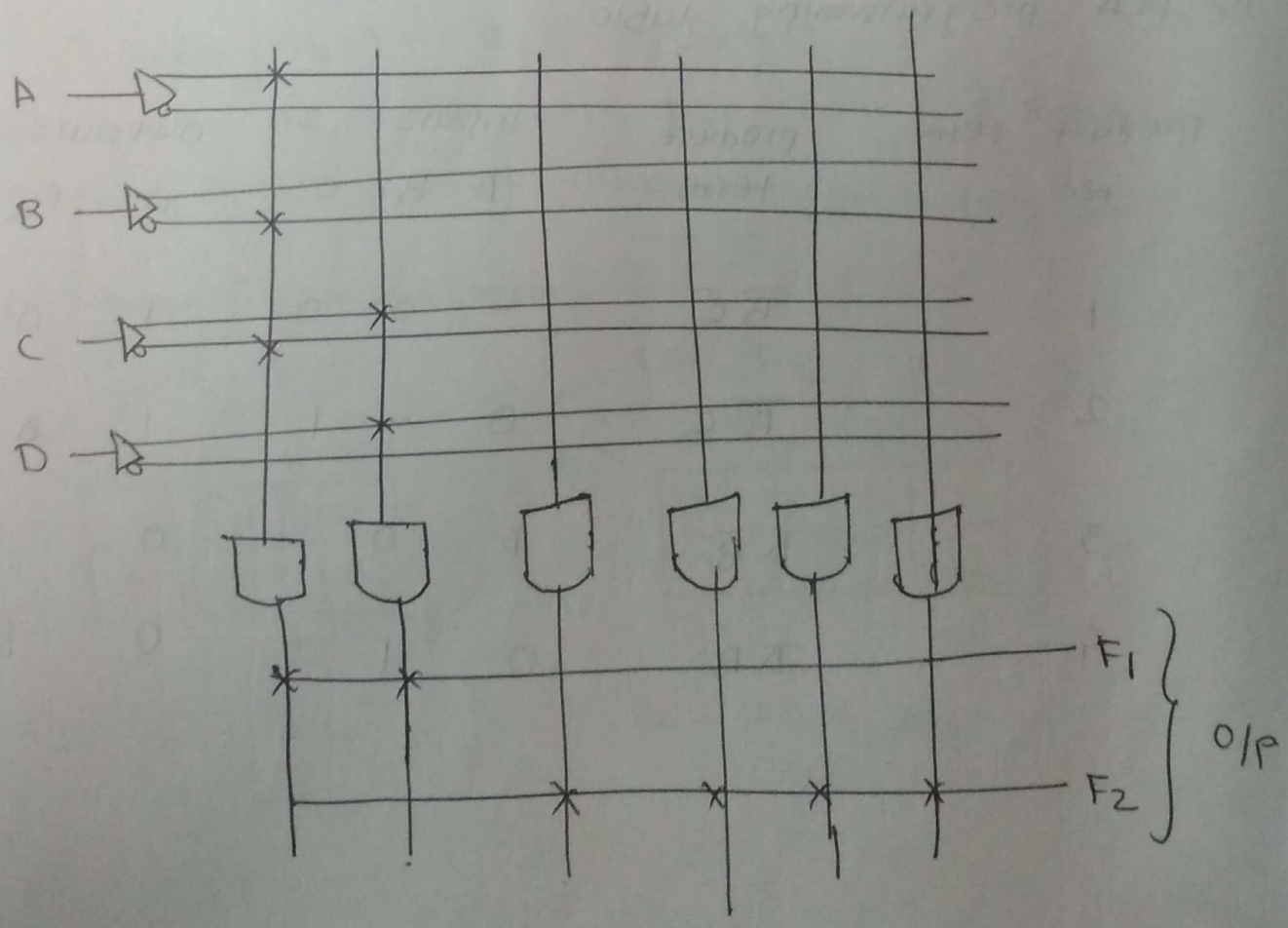
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ii) Prepare PLA programming table →

Product term no.	Product term	I/p A	B	C	D	O/p F_1	F_2
1	$A\bar{B}\bar{C}$	1	0	0	–	1	0
2	CD	–	–	1	1	1	0
3	$\bar{A}B\bar{C}$	0	1	0	–	0	1
4	$\bar{A}CD$	0	–	1	1	0	1
5	ABC	1	1	1	0	0	1
6	$AC\bar{D}$	1	–	1	0	0	1

iii) Logic circuit :-



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3) A combinational circuit is defined by the following functions. Implement this circuit with PLA having 3, 4 product terms & 2 outputs

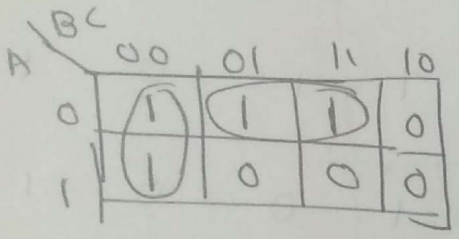
(8 Marks - Dec = 2014)

$F_1(A,B,C) = \Sigma m(0,1,3,4)$

$F_2(A,B,C) = \Sigma m(1,2,3,4,5)$

⇒ i) K-map & simplification.

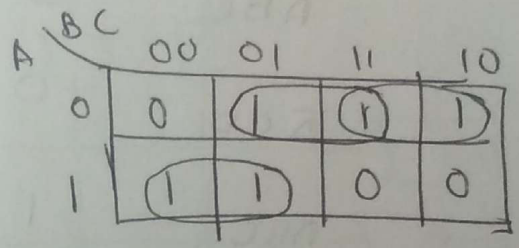
FOR F_1



$F_1 = \overline{B}\,\overline{C} + \overline{A}C$

(1) (2)

FOR F_2



$F_2 = A\overline{B} + \overline{A}C + \overline{A}B$

(3) (2) (4)

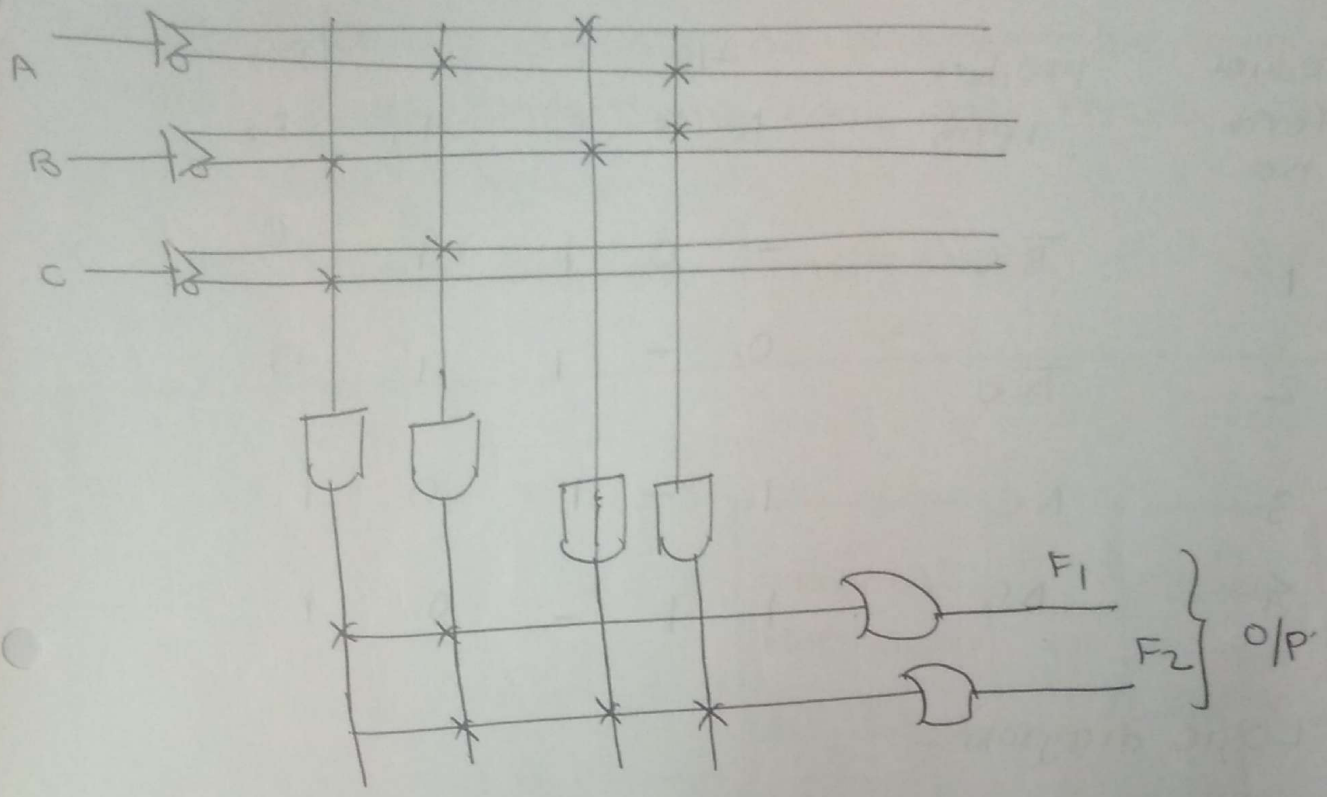
ii) PLA programming table :-

product term No.	product term	Inputs A	B	C	Outputs F_1	F_2
1	$\overline{B}\,\overline{C}$	–	0	0	1	0
2	$\overline{A}C$	0	–	1	1	1
3	$A\overline{B}$	1	0	–	0	1
4	$\overline{A}B$	0	1	–	0	1

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ii) Logic diagram



4) A combinational circuit is defined by a function ~~F1=Σm~~

$F_1 = \Sigma m(1,3,5)$ $F_2 = \Sigma m(5,6,7)$

Implement the circuit with PLA having 3 inputs, 3 product terms & two outputs. (8M - May 2015)

5) i) k-map & simplification.

For F_1

A \ BC	00	01	11	10
0	0	1	1	0
1	0	1	0	0

$F_1 = \overline{B}C + \overline{A}C$
(1) (2)

For F_2

A \ BC	00	01	11	10
0	0	0	0	0
1	0	1	1	1

$F_2 = AC + AB$
(3) (4)

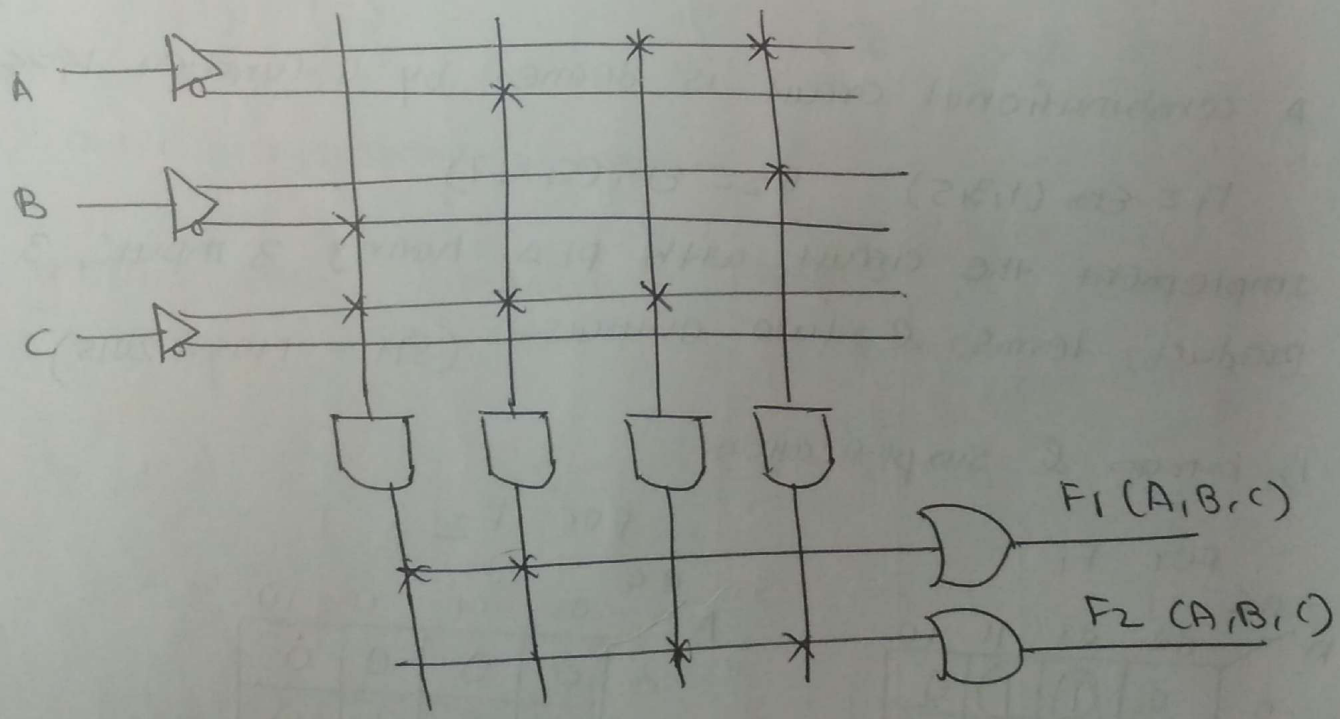
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ii) PLA programming table :-

product term no.	product term	I/p A	I/p B	I/p C	O/p F_1	O/p F_2
1	$\overline{B}C$	–	0	1	1	0
2	$\overline{A}C$	0	–	1	1	0
3	AC	1	–	1	0	1
4	AB	1	1	–	0	1

iii) Logic diagram :-



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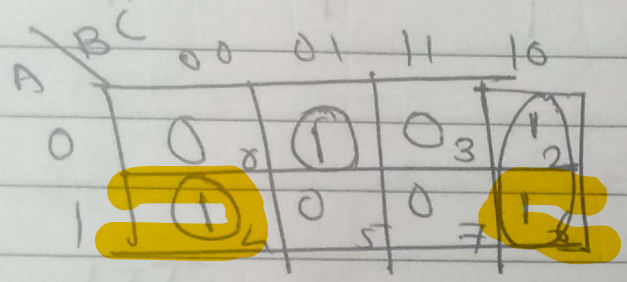
4 (b) Implement using PLA

$F_1(A,B,C) = \Sigma m(1,2,4,6)$

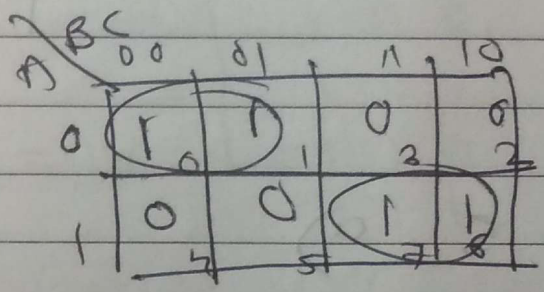
$F_2(A,B,C) = \Sigma m(0,1,6,7)$

$\Rightarrow$

F_1



F_2



$\overline{F_1} = \overline{A}\,\overline{B}C + B\overline{C} + A\overline{B}\,\overline{C}$

(1) (2) or (3)

$= \overline{A}\,\overline{B}C + \overline{A}B\overline{C} + A\overline{C}$

$F_2 = \overline{A}\,\overline{B} + AB$

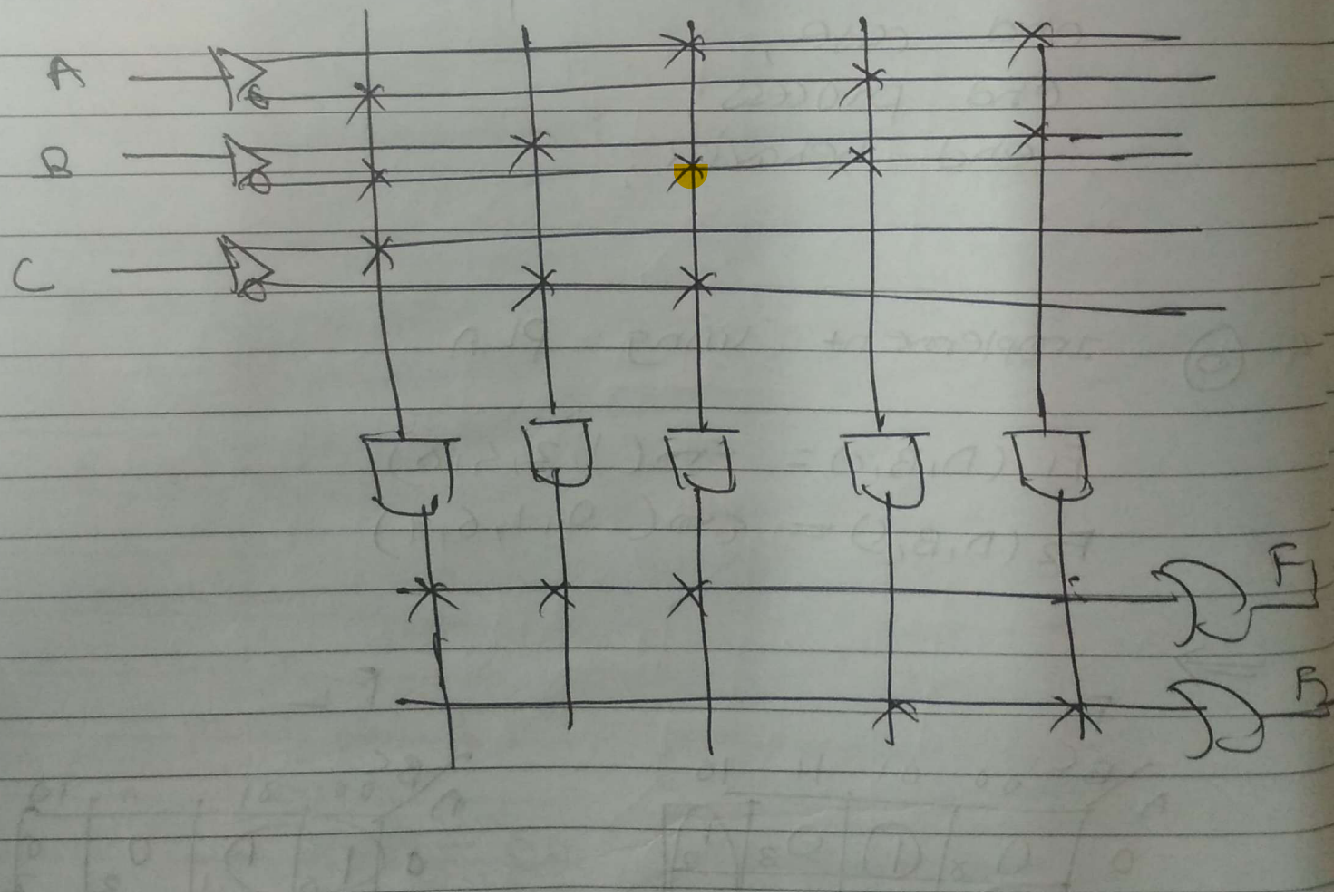
(4) (5)

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PLA table

P.I. No.	P.T	A	B	C	F_1	F_2
1	$\bar{A}\bar{B}C$	0	0	1	1	—
2	$B\bar{C}$	—	1	0	1	—
3	$A\bar{B}C$	1	0	0	1	—
4	$\bar{A}\bar{B}$	0	0	—	—	1
5	AB	1	1	—	—	1



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Programmable Array Logic (PAL) :-

i) PAL is the most commonly used type of PLD. It is a programmable array of logic gates.

ii) The special feature of PAL is that it has a programmable AND array & a fixed OR array.

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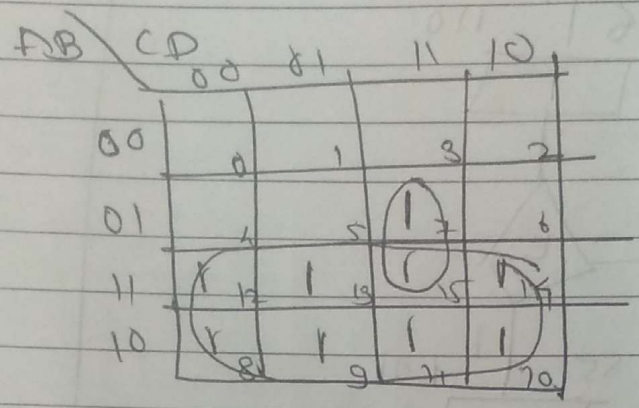
Q3 (b) Implement using PAL

$F_1(A,B,C,D) = \Sigma m(7,8,9,10,11,12,13,14,15)$

$F_2(A,B,C,D) = \Sigma m(2,12,13)$

$\Rightarrow$

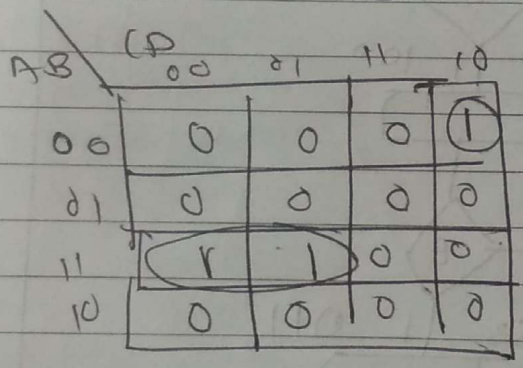
F1



$F1 = BCD + A$

① ②

F_2



$F_2 = \bar{A}\bar{B}C\bar{D} + AB\bar{C}$

③ ④

pr. t. no	product term	A	B	CD
1	BCD	–	1	1 1
2	A	1	–	– –

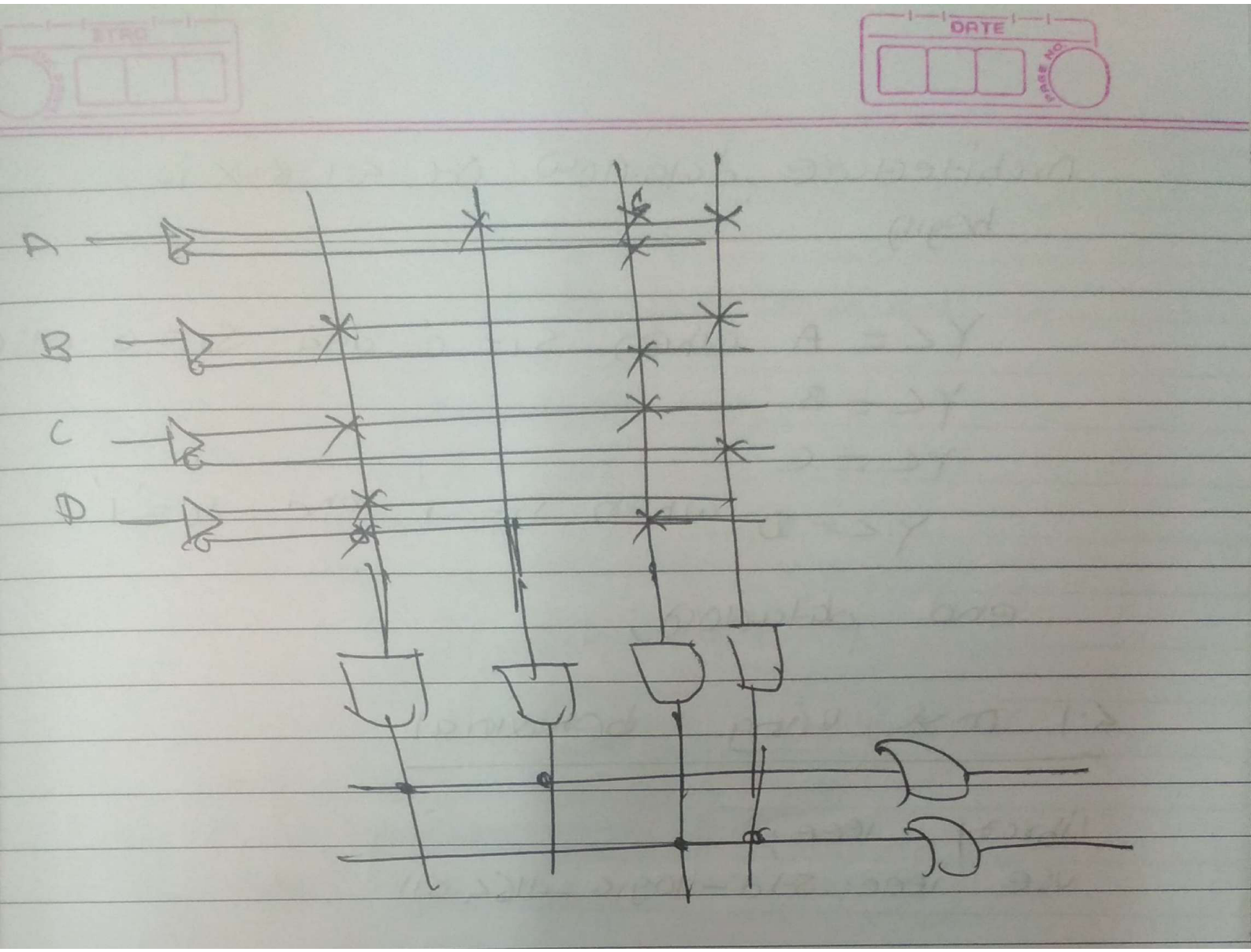
$F_1 = A + BCD$

3	$\bar{A}\bar{B}C\bar{D}$	0	0	1 0
4	$AB\bar{C}$	1	1	0 –

$F_2 = \bar{A}\bar{B}C\bar{D} + AB\bar{C}$

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1) Design a BCD to excess-3 code converter & implement it using PAL. (8M – (Dec 2013, 2014)

⇒ i) write truth table

Decimal	BCD i/p				Excess-3 i/p			
	D_3	D_2	D_1	D_0	E_3	E_2	E_1	E_0
0	0	0	0	0	0	0	1	1
1	0	0	0	1	0	1	0	0
2	0	0	1	0	0	1	0	1
3	0	0	1	1	0	1	1	0
4	0	1	0	0	0	1	1	1
5	0	1	0	1	1	0	0	0
6	0	1	1	0	1	0	0	1
7	0	1	1	1	1	0	1	0
8	1	0	0	0	1	0	1	1
9	1	0	0	1	1	1	0	0

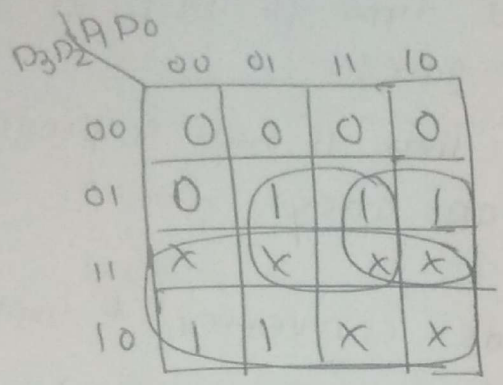
BCD no. will count from 0→9 After 10th no. all condition's are don't care

ii) K-map & simplified expression-:

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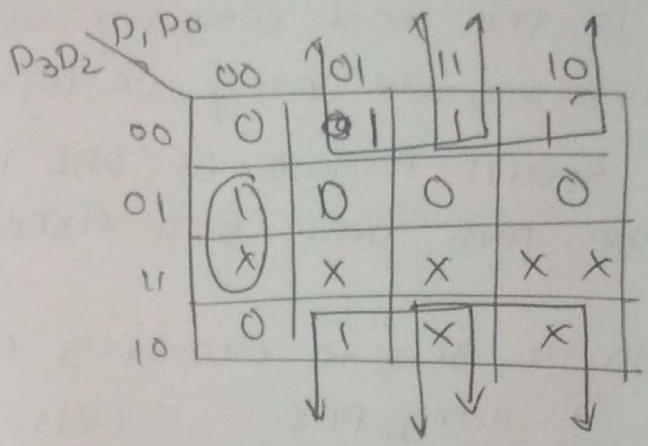
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FOR E_3



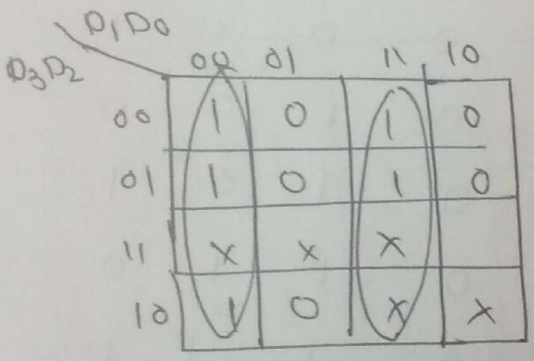
$E_3 = D_3 + D_2 \cdot D_0 + D_2 \cdot D_1$
(4) (2) (3)

FOR E_2



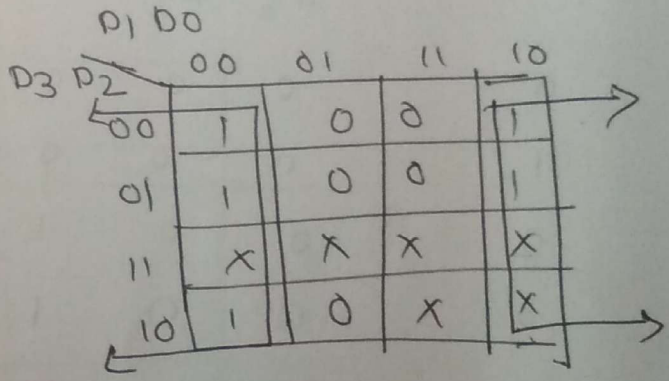
$E_2 = D_2\bar{D}_1\bar{D}_0 + \bar{D}_2 \cdot D_0 + D_1\bar{D}_2$
(1) (5) (6)

FOR E_1



$E_1 = \bar{D}_1\bar{D}_0 + D_1D_0$
(7) (8)

FOR E_0



$E_0 = \bar{D}_0$
(9)

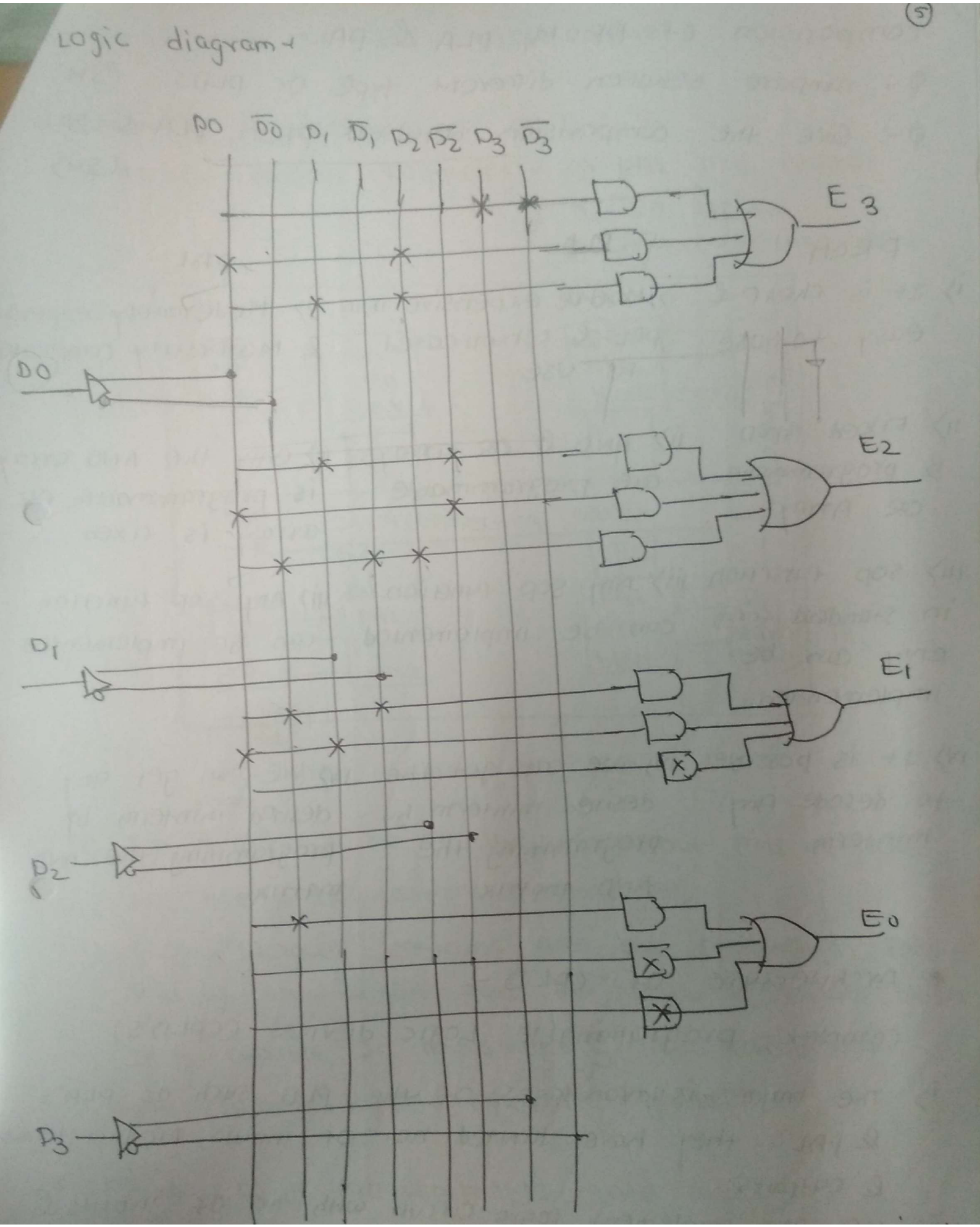
iii) PAL programming table →

product term no.	product term	AND i/p $D_3\ D_2\ D_1\ D_0$	o/p
1	D_3	1 – – –	$E_3 = D_3 + D_2D_0 + D_2D_1$
2	D_2D_0	– 1 – 1	
3	D_2D_1	~~0~~ 1 1 ~~0~~	
4	$D_2\bar{D}_1\bar{D}_0$	– 1 0 0	$E_2 = D_2\bar{D}_1\bar{D}_0 + \bar{D}_2D_0 + D_1\bar{D}_2$
5	$\bar{D}_2D_0$	– 0 – 1	
6	$D_1\bar{D}_2$	– 0 1 –	
7	$\bar{D}_1\bar{D}_0$	– – 0 0	$E_1 = \bar{D}_1\bar{D}_0 + D_1D_0$
8	D_1D_0	– – 1 1	
9	$\bar{D}_0$	– – – 0	$\rightarrow \bar{D}_0 = E_0$

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Logic diagram :



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PROM	PLA	PAL
It is cheap and easy to use.	It is expensive than PAL and PROM and complicated to use.	Moderately expensive and moderately complicated.
There are fixed AND arrays and programmable OR arrays.	AND and OR arrays are programmable	Only the AND array is programmable. OR array is fixed.
SOP functions in the standard form only can be implemented.	Any SOP function can be implemented	Any SOP function can be implemented.
It is possible to decode any minterm.	We can get any desired minterm by programming the AND matrix.	We can get any desired minterm by programming the AND matrix.

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